# Notes: Ackerberg, Caves, and Frazer (Econometrica, 2015)

**Identification Properties of Recent Production Function Estimators**

## Contents

# 1 Big picture

- **Question.** Do the popular proxy-variable estimators of production functions—especially Olley–Pakes (OP) and Levinsohn–Petrin (LP)—actually identify the labor coefficient the way the literature often claims?
- **Setting.** Firms choose inputs while privately observing productivity. The econometrician observes output and inputs, but not productivity. OP uses investment as a proxy for productivity; LP uses an intermediate input such as materials.
- **Core challenge.** In OP and LP, the first stage tries to partial out unobserved productivity with a nonparametric control function and then identify labor from the remaining variation. ACF show that, under natural timing and choice rules, labor is often a deterministic function of exactly the variables entering that control function.
- **Main critique.** This creates a *functional dependence* problem: after conditioning nonparametrically on capital and the proxy variable, there may be no independent variation left in labor, so the first-stage estimate of the labor coefficient is not identified.
- **Main theoretical point.** The problem is not that OP/LP always fail numerically. They usually still return a number. The point is conceptual: unless the data come from a very specific set of data-generating processes, that number is not a consistent estimate of the labor coefficient.
- **Main contribution.** ACF propose a modified proxy-variable estimator that inverts a *conditional* input demand function, for example materials demand conditional on labor, instead of the usual unconditional one.
- **Key trade-off.** Their first stage no longer tries to identify labor. Instead, it only nets out the unanticipated error term. All structural coefficients are then identified in the second stage.
- **Why this matters.** The modification is small in appearance but large in consequence. It avoids the functional dependence problem and allows more realistic environments, including dynamic labor, firm-specific unobserved wage shocks, and labor chosen earlier than other variable inputs.
- **Quantitative evidence.** In Monte Carlo designs favorable to ACF, LP can be severely biased while ACF remains close to the truth. In designs favorable to LP, both work, but LP is more efficient. Under measurement error in materials, ACF is generally less sensitive than LP.
- **Economic takeaway.** The paper is not merely a technical correction. It changes how one should think about identification in proxy-variable production function estimation: the critical issue is not whether a proxy exists, but whether the proxy-based control function leaves any independent variation in the flexible input coefficient one wants to estimate.

# 2 Data and measurement (key objects)

## 2.1 What is observed in the OP/LP/ACF setup

- The econometrician observes firm-level output and inputs in a panel with many firms and fixed time dimension.
- In the simplest value-added specification, the key observed variables are:

  - $y_{it}$: log output,
  - $k_{it}$: log capital,
  - $l_{it}$: log labor,
  - and either $i_{it}$ (investment) or $m_{it}$ (materials/intermediate input) as the proxy variable.
- The econometrician does *not* observe:
  - $\omega_{it}$: productivity known or predictable by the firm when inputs are chosen,
  - $\varepsilon_{it}$: the innovation or measurement-error component that is not known when the firm chooses inputs.

## 2.2 The core econometric object

The benchmark value-added production function is

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \varepsilon_{it}. \tag{1}$$

- **Interpretation.** $\beta_k$ and $\beta_l$ are the capital and labor elasticities the researcher wants to estimate.
- **Problem.** If firms choose inputs after observing $\omega_{it}$, then $k_{it}$ and $l_{it}$ are endogenous and OLS is biased.

## 2.3 Why OP and LP need proxy variables

- OP and LP both rely on the idea that a firm's input choice reveals information about productivity.
- If an input demand function is strictly monotone in productivity, the econometrician can invert it and write productivity as a function of observables.
- Then one can plug that inversion into the production function and control nonparametrically for productivity.

## 2.4 What is and is not identified from data alone

- The paper's message is that one should not confuse *numerical estimation* with *identification*.
- OP and LP can almost always be run in finite samples and return estimates.
- But whether the estimated labor coefficient is truly identified depends on the structure of input timing, optimization error, and which unobservables enter which input demand equations.

## 2.5 Measurement issues

- The theory is easiest when inputs and output are measured in comparable physical units across firms.
- In practice, researchers often observe monetary values, not quantities.
- Then one must worry about firm-specific price differences, input-quality differences, and market power in output and input markets.

- ACF explain that if equivalent-price variation is observed, one can divide by prices and recover physical units. If prices are not observed, additional structure is needed, and the standard OP/LP/ACF machinery is no longer enough by itself.

### 2.6 Monte Carlo objects

The quantitative section uses a simulated panel designed to mimic broad features of plant data used by LP:

- 1,000 firms,
- 10 periods,
- a Leontief-in-materials technology,
- true elasticities $\beta_k = 0.4$, $\beta_l = 0.6$, and $\beta_m = 1$,
- persistent productivity,
- dynamic capital with convex adjustment costs,
- and alternative assumptions about labor timing, wage shocks, optimization error, and measurement error in materials.

These DGPs are not meant to estimate a real industry. They are designed to show *when* different estimators work and *how* they break under misspecification.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 The basic endogeneity problem

Start from the Cobb–Douglas value-added production function:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \varepsilon_{it}. \tag{1 revisited}$$

- $\omega_{it}$ is the part of productivity the firm knows when choosing inputs.
- $\varepsilon_{it}$ is the unexpected shock or measurement error.
- Because firms choose inputs using $\omega_{it}$, input choices are correlated with the unobserved term, so OLS is not valid.

ACF review three broad traditions before OP/LP:

- fixed effects and dynamic panel methods,
- estimation via first-order conditions,
- and input-price IV approaches.

Their point is that OP/LP relax some assumptions of these older approaches, but pay for that flexibility with new assumptions, especially timing, scalar unobservability, and monotonicity.

### 3.2 Olley–Pakes (OP)

OP assume:

- firms observe current productivity before choosing variable inputs,
- productivity follows a first-order Markov process,
- capital is dynamic and predetermined,
- labor is static and chosen contemporaneously,
- investment is a strictly increasing function of capital and productivity.

Formally, the investment policy is

$$i_{it} = f_t(k_{it}, \omega_{it}), \tag{2}$$

and strict monotonicity lets the econometrician invert it:

$$\omega_{it} = f_t^{-1}(k_{it}, i_{it}). \tag{3}$$

Substituting this into the production function gives the OP first stage:

$$y_{it} = \beta_l l_{it} + \Phi_t(k_{it}, i_{it}) + \varepsilon_{it}. \tag{4}$$

The first-stage moment condition is

$$E\big[y_{it} - \beta_l l_{it} - \Phi_t(k_{it}, i_{it}) \mid I_{it}\big] = 0. \tag{5}$$

Then productivity is assumed to evolve as

$$\omega_{it} = g(\omega_{it-1}) + \xi_{it}, \qquad E[\xi_{it} \mid I_{it-1}] = 0, \tag{1}$$

which implies a second-stage moment of the form

$$\begin{aligned} E[\xi_{it} + \varepsilon_{it} \mid I_{it-1}] = E\Big[&y_{it} - \beta_0 - \beta_k k_{it} - \beta_l l_{it} \\ &- g\big(\Phi_{t-1}(k_{it-1}, i_{it-1}) - \beta_0 - \beta_k k_{it-1}\big) \mid I_{it-1}\Big] = 0. \end{aligned} \tag{8}$$

**Interpretation.**

- OP identify labor in the *first* stage and capital in the *second* stage.
- The whole strategy depends on the first stage leaving independent variation in labor after conditioning on $k$ and $i$.

### 3.3 Levinsohn–Petrin (LP)

LP replace investment with a static intermediate input. Their gross-output specification is

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \omega_{it} + \varepsilon_{it}. \tag{9}$$

They assume the intermediate input demand is

$$m_{it} = f_t(k_{it}, \omega_{it}), \tag{10}$$

with strict monotonicity, so that

$$\omega_{it} = f_t^{-1}(k_{it}, m_{it}). \tag{2}$$

The LP first stage becomes

$$y_{it} = \beta_l l_{it} + \Phi_t(k_{it}, m_{it}) + \varepsilon_{it}, \tag{12}$$

with moment condition

$$E\big[y_{it} - \beta_l l_{it} - \Phi_t(k_{it}, m_{it}) \mid I_{it}\big] = 0. \tag{13}$$

Then the second stage is

$$\begin{aligned} E[\xi_{it} + \varepsilon_{it} \mid I_{it-1}] = E\Big[&y_{it} - \beta_0 - \beta_k k_{it} - \beta_l l_{it} - \beta_m m_{it} \\ &- g\big(\Phi_{t-1}(k_{it-1}, m_{it-1}) - \beta_0 - \beta_k k_{it-1} - \beta_m m_{it-1}\big) \mid I_{it-1}\Big] = 0. \end{aligned} \tag{14}$$

**Interpretation.**

- LP are often easier to implement because materials are usually smoother and less zero-valued than investment.
- LP also allow unobserved heterogeneity in capital adjustment costs that would violate OP's scalar-unobservable assumption for investment.
- But LP still identify labor in the first stage, which is exactly where ACF locate the problem.

## 3.4 The functional dependence critique

ACF's cleanest intuition comes from a parametric Cobb–Douglas FOC for materials. If firms are price takers, the first-order condition for materials implies that after solving for productivity and substituting back into the production function, labor drops out. In the parametric illustration, the first-stage equation becomes

$$y_{it} = \ln\left(\frac{p^m}{\beta_m p^y}\right) + \ln(\beta_m) + m_{it} + \varepsilon_{it}, \tag{15}$$

so $\beta_l$ is not present at all.

The nonparametric version uses the Robinson partially linear identification condition:

$$E\Big[\big(l_{it} - E[l_{it} \mid k_{it}, m_{it}, t]\big)\big(l_{it} - E[l_{it} \mid k_{it}, m_{it}, t]\big)'\Big] \text{ must be positive definite.} \tag{16}$$

If labor is functionally dependent on the variables inside the control function, then this condition fails.

The simplest data-generating process for labor is

$$l_{it} = h_t(k_{it}, \omega_{it}). \tag{17}$$

But since LP also use $m_{it} = f_t(k_{it}, \omega_{it})$, inversion implies labor can be written as a function of $k_{it}$ and $m_{it}$ alone. Then there is no variation left in labor after controlling for $\Phi_t(k_{it}, m_{it})$.

**Interpretation.**

- The criticism is not that the first-stage regression cannot be run.
- It is that the nonparametric term is soaking up exactly the same variation that is supposed to identify labor.
- In applied work, one may still get a coefficient because the polynomial approximation is imperfect or because the sample is finite. But that does not rescue identification.

### 3.5 When can OP/LP first-stage labor identification survive?

ACF argue that the OP/LP first stage identifies labor only under three narrow classes of DGPs:

- **(1) i.i.d. optimization error in labor only.** Labor deviates from its optimum by independent noise, but the proxy input does not. Then labor retains independent variation.
- **(2) i.i.d. labor-price or output-price shocks arriving after the proxy input is chosen but before labor is chosen.** These shocks shift labor without shifting the proxy input, again restoring variation.
- **(3) OP-only timing case.** Labor is chosen at an intermediate subperiod $t-b$ based on $\omega_{it-b}$, investment is chosen later using $\omega_{it}$, and labor has no dynamic effects. Then labor need not be a deterministic function of $(k, i)$.

ACF's judgment is that these cases are special and often implausible as generic justifications for first-stage identification in empirical work.

### 3.6 ACF's alternative procedure: conditional input demand

The alternative value-added production function is the same as before:

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \varepsilon_{it}. \tag{23}$$

But the timing and proxy assumptions change.

- Labor may have dynamic implications and can be chosen at $t$, $t-1$, or some intermediate subperiod $t-b$.
- The materials demand function is assumed to be *conditional on labor*:

$$m_{it} = f_t(k_{it}, l_{it}, \omega_{it}). \tag{24}$$

With strict monotonicity, this implies

$$\omega_{it} = f_t^{-1}(k_{it}, l_{it}, m_{it}). \tag{3}$$

Substituting into production gives

$$y_{it} = \Phi_t(k_{it}, l_{it}, m_{it}) + \varepsilon_{it}. \tag{25}$$

The first-stage moment is now simply

$$E\big[y_{it} - \Phi_t(k_{it}, l_{it}, m_{it}) \mid I_{it}\big] = 0. \tag{26}$$

**Key point.** This first stage does *not* identify $\beta_l$. That is intentional.

The second stage identifies all coefficients:

$$\begin{aligned} E[\xi_{it} + \varepsilon_{it} \mid I_{it-1}] = E\Big[ & y_{it} - \beta_0 - \beta_k k_{it} - \beta_l l_{it} \\ & - g\big(\Phi_{t-1}(k_{it-1}, l_{it-1}, m_{it-1}) - \beta_0 - \beta_k k_{it-1} - \beta_l l_{it-1}\big) \mid I_{it-1}\Big] = 0. \end{aligned} \tag{27}$$

If productivity is AR(1), $\omega_{it} = \rho\omega_{it-1} + \xi_{it}$, a natural unconditional version is

$$\begin{aligned} E\Big[ & \big(y_{it} - \beta_0 - \beta_k k_{it} - \beta_l l_{it} - \rho\left[\Phi_{t-1}(k_{it-1}, l_{it-1}, m_{it-1}) - \beta_0 - \beta_k k_{it-1} - \beta_l l_{it-1}\right]\big) \\ & \otimes \begin{pmatrix} 1 \\ k_{it} \\ l_{it-1} \\ \Phi_{t-1}(k_{it-1}, l_{it-1}, m_{it-1}) \end{pmatrix} \Big] = 0. \end{aligned} \tag{28}$$

**Interpretation.**

- Conditioning on labor in the proxy inversion means that labor is no longer competing with the control function for variation in the first stage.
- Identification of labor is pushed to the second stage, where lagged instruments and the Markov law of motion for productivity do the work.
- The price is lower efficiency when the more restrictive LP assumptions actually hold.

### 3.7 Why the conditional approach is more general

Relative to standard LP, ACF's approach can accommodate:

- serially correlated unobserved wage shocks,
- dynamic labor with adjustment costs,
- labor chosen before materials with a different information set.

The basic logic is simple: conditional on labor, the materials demand function need not depend on the omitted labor-side disturbance, while the unconditional materials demand function generally does.

What ACF still *cannot* allow is unobserved firm-level shocks to the cost of materials themselves. If such shocks are omitted, neither conditional nor unconditional inversion is valid because the scalar-unobservable restriction fails.

### 3.8 Investment-function version of the ACF idea

ACF also show how to implement the same idea with investment instead of materials. One replaces OP's unconditional investment demand function with a conditional one,

$$i_{it} = f_t(k_{it}, l_{it}, \omega_{it}), \tag{4}$$

and again uses a first stage that absorbs $\beta_l$ into the control function.

**Interpretation.**

- This avoids OP's first-stage labor-identification problem for the same reason.
- But it is less robust than the materials version when there are omitted shocks affecting investment, such as firm-specific capital adjustment costs or investment prices.

### 3.9 Relation to Wooldridge (2009)

Wooldridge proposes stacking the first and second stage moments and estimating them jointly. For LP, the stacked moments are built around an *unconditional* materials demand function.

ACF agree that joint estimation helps avoid the narrow first-stage identification issue. But they stress that Wooldridge's formulation still imposes the unconditional inversion. Thus it does not accommodate the same broader set of DGPs as the ACF conditional approach.

Conceptually:

- Wooldridge solves the functional dependence problem by moving to joint estimation.
- ACF solve it by changing the object being inverted: conditional rather than unconditional input demand.

### 3.10 Relation to dynamic panel estimators

ACF compare their approach with dynamic panel methods. Suppose productivity is linear AR(1):

$$y_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \varepsilon_{it}, \tag{32}$$

with $\omega_{it} = \rho\omega_{it-1} + \xi_{it}$. Then one can quasi-difference:

$$\begin{aligned} y_{it} - \rho y_{it-1} = \beta_0(1-\rho) + \beta_k(k_{it} - \rho k_{it-1}) \\ + \beta_l(l_{it} - \rho l_{it-1}) + \xi_{it} + (\varepsilon_{it} - \rho\varepsilon_{it-1}). \end{aligned} \tag{33}$$

The resulting moment condition is

$$E\left[\xi_{it} + (\varepsilon_{it} - \rho\varepsilon_{it-1}) \mid I_{it-1}\right] = 0. \tag{34}$$

**Interpretation.**

- Dynamic panel methods do not require scalar-unobservable proxy inversions.
- But they typically require stronger parametric assumptions on the productivity process, especially linearity.
- ACF's point is not that one method dominates. It is that the identifying assumptions differ, and researchers should be explicit about which set they find more plausible.

### 3.11 Gross output and units of measure

ACF emphasize that their alternative procedure is developed for *value-added* production functions. They caution against applying the same logic directly to general gross-output production functions with non-Leontief materials because identification then fails without extra restrictions.

They also stress that when output or inputs are measured in monetary values, firm-specific price variation can break comparability across firms. In those cases one needs either observed prices or further structure from demand or input-supply models.

# 4 Identification and instruments (what makes the estimates credible)

## 4.1 What identifies OP and LP when their assumptions are true

Under the OP/LP logic, identification comes from two steps:

- **Step 1:** monotonicity lets the econometrician invert investment or materials demand to control for productivity.
- **Step 2:** the Markov evolution of productivity and lagged predetermined inputs identify the remaining coefficients.

This logic is elegant, but it only works if the first stage really leaves independent variation in labor.

## 4.2 Why ACF say first-stage labor identification is generally not credible

The paper's core identification argument is negative but powerful.

- If labor is chosen using the same information and under the same omitted state variables as the proxy input, then labor becomes a function of the proxy variables.
- Once one conditions on those proxy variables nonparametrically, labor is absorbed into the control function.
- Therefore the first-stage moment cannot identify the labor elasticity except under highly specific timing or noise assumptions.

The point is especially strong because it does not rely on small-sample pathologies. It is an identification argument.

## 4.3 What identifies ACF's alternative estimator

In ACF, the first stage identifies only the composite object

$$\Phi_t(k_{it}, l_{it}, m_{it}) = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \omega_{it}, \tag{5}$$

not the coefficients separately.

Identification of $(\beta_0, \beta_k, \beta_l)$ then comes from:

- the first-order Markov law of motion for productivity,
- lagged instruments such as $k_{it}$, $l_{it-1}$, and lagged control functions,
- and the fact that the current innovation $\xi_{it}$ is orthogonal to time $t-1$ information.

This is why the ACF approach is conceptually cleaner: the first stage is used only to purge the unanticipated error, while coefficient identification is moved to where the orthogonality conditions genuinely live.

### 4.4 Why the ACF approach can be more general

Using a conditional materials demand function means that unobserved labor-side disturbances do not necessarily contaminate the proxy inversion. This is what lets ACF allow:

- dynamic labor,
- serially correlated unobserved wages,
- and labor chosen earlier than other variable inputs.

In contrast:

- OP cannot handle omitted shocks to capital costs or adjustment costs because these enter investment demand,
- LP cannot handle omitted serially correlated labor-cost shocks when using unconditional materials demand,
- Wooldridge's stacked LP moments avoid the narrow first-stage issue but still inherit the unconditional inversion restriction.

### 4.5 What makes the critique persuasive

Several features make the paper especially credible.

- The critique is derived both parametrically and nonparametrically.
- The authors do not claim OP/LP are always useless; they carefully characterize the narrow DGPs under which those methods can still work.
- Their proposed estimator is not a radical replacement. It stays very close to the spirit of OP/LP, which makes the comparison transparent.
- They benchmark the estimator against favorable and unfavorable DGPs in Monte Carlo, rather than only showing cases where their own method wins.

### 4.6 Main limitations

- The ACF estimator still requires a scalar unobservable and strict monotonicity.
- It is naturally suited to value-added production functions, not unrestricted gross-output ones.
- It still cannot handle omitted firm-level shocks to materials prices or demand conditions that enter materials demand.
- Dynamic panel alternatives remain attractive if one is willing to impose a more parametric productivity process.

- In practice, the approach can be computationally heavier than simple LP implementations, especially with flexible nonparametric approximations and many moments.

### 4.7 Relation to the literature

- Relative to fixed effects and dynamic panels, ACF keep the proxy-variable insight but rethink what is actually identified.
- Relative to OP, LP already improved the proxy variable by using a non-dynamic input. ACF's contribution is to show that even LP's first stage may not identify labor.
- Relative to Wooldridge, ACF highlight that the distinction is not only one-step versus two-step estimation, but unconditional versus conditional inversion.
- Later work on gross-output identification, especially Gandhi–Navarro–Rivers, takes ACF's warning seriously and develops additional restrictions for cases where materials enter production directly.

## 5 Tables and figures

### 5.1 A note on visuals

- Unlike many empirical papers, this paper has essentially **no central main-text figures** driving the argument.
- The quantitative evidence is concentrated in the Monte Carlo tables.
- So the right way to read the paper is: first understand the identification logic, then use the tables as disciplined numerical illustrations of when each estimator works or fails.

### 5.2 Table I: ACF versus LP under three DGPs

**Read:** Table I is the key quantitative exhibit. It compares the ACF estimator with LP across three data-generating processes and several levels of measurement error in materials.

- **DGP1 (favorable to ACF):** serially correlated wage shocks plus labor chosen at time $t-b$.
  - With no materials measurement error, ACF recovers the truth almost exactly: $\hat{\beta}_l = 0.600$ and $\hat{\beta}_k = 0.399$.
  - LP fails badly even with no measurement error: $\hat{\beta}_l = 0.000$ and $\hat{\beta}_k = 1.121$.
  - As measurement error rises to 0.5, ACF remains relatively stable ($\hat{\beta}_l = 0.629$, $\hat{\beta}_k = 0.405$), while LP swings sharply ($\hat{\beta}_l = 0.754$, $\hat{\beta}_k = 0.291$).
- **DGP2 (favorable to LP):** optimization error in labor only.
  - With no measurement error, both procedures work: ACF gives about $(0.600, 0.400)$ and LP gives about $(0.600, 0.399)$ for $(\beta_l, \beta_k)$.
  - LP is much more precise here: its standard deviations are around 0.003 and 0.013, compared with ACF's roughly 0.009 and 0.016.

  - This is exactly the trade-off ACF emphasize: their procedure is more general, but less efficient when LP's restrictive assumptions are actually correct.

- **DGP3 (favorable to neither):** optimization error in labor plus serially correlated wage shocks and labor chosen at $t - b$.
  - Neither estimator is exactly consistent.
  - But ACF stays closer to the truth and is less sensitive to added measurement error. For example, with zero measurement error ACF gives $(0.596, 0.406)$ while LP gives $(0.473, 0.588)$.

**Why this table matters.**

- It makes clear that the paper is not claiming universal dominance of ACF.
- Instead, it shows a disciplined bias-efficiency trade-off:
  - LP is efficient when its assumptions hold,
  - ACF is more robust when those assumptions do not,
  - and ACF is often less fragile to measurement error in the proxy variable.

TABLE I

MONTE CARLO RESULTS[a]

| Meas. Error | ACF $\beta_l$ Coef. | ACF $\beta_l$ Std. Dev. | ACF $\beta_k$ Coef. | ACF $\beta_k$ Std. Dev. | LP $\beta_l$ Coef. | LP $\beta_l$ Std. Dev. | LP $\beta_k$ Coef. | LP $\beta_k$ Std. Dev. |
|---|---|---|---|---|---|---|---|---|
| *DGP1—Serially Correlated Wages and Labor Set at Time $t-b$* | | | | | | | | |
| 0.0 | 0.600 | 0.009 | 0.399 | 0.015 | 0.000 | 0.005 | 1.121 | 0.028 |
| 0.1 | 0.596 | 0.009 | 0.428 | 0.015 | 0.417 | 0.009 | 0.668 | 0.019 |
| 0.2 | 0.602 | 0.010 | 0.427 | 0.015 | 0.579 | 0.008 | 0.488 | 0.015 |
| 0.5 | 0.629 | 0.010 | 0.405 | 0.015 | 0.754 | 0.007 | 0.291 | 0.012 |
| *DGP2—Optimization Error in Labor* | | | | | | | | |
| 0.0 | 0.600 | 0.009 | 0.400 | 0.016 | 0.600 | 0.003 | 0.399 | 0.013 |
| 0.1 | 0.604 | 0.010 | 0.408 | 0.016 | 0.677 | 0.003 | 0.332 | 0.011 |
| 0.2 | 0.608 | 0.011 | 0.410 | 0.015 | 0.725 | 0.003 | 0.289 | 0.010 |
| 0.5 | 0.620 | 0.013 | 0.405 | 0.017 | 0.797 | 0.003 | 0.220 | 0.010 |
| *DGP3—Optimization Error in Labor and Serially Correlated Wages and Labor Set at Time $t-b$ (DGP1 plus DGP2)* | | | | | | | | |
| 0.0 | 0.596 | 0.006 | 0.406 | 0.014 | 0.473 | 0.003 | 0.588 | 0.016 |
| 0.1 | 0.598 | 0.006 | 0.422 | 0.013 | 0.543 | 0.004 | 0.522 | 0.014 |
| 0.2 | 0.601 | 0.006 | 0.428 | 0.012 | 0.592 | 0.004 | 0.473 | 0.012 |
| 0.5 | 0.609 | 0.007 | 0.431 | 0.013 | 0.677 | 0.005 | 0.386 | 0.012 |

[a]1000 replications. True values of $\beta_l$ and $\beta_k$ are 0.6 and 0.4, respectively. Standard deviations reported are of parameter estimates across the 1000 replications.

TABLE II

MONTE CARLO RESULTS[a]

| Meas. Error | OP $\beta_l$ Coef. | OP $\beta_l$ Std. Dev. | OP $\beta_k$ Coef. | OP $\beta_k$ Std. Dev. |
|---|---|---|---|---|
| *DGP1—Serially Correlated Wages and Labor Set at Time $t-b$* | | | | |
| 0.0 | 0.824 | 0.002 | 0.188 | 0.004 |
| 0.1 | 0.836 | 0.002 | 0.179 | 0.004 |
| 0.2 | 0.845 | 0.002 | 0.171 | 0.003 |
| 0.5 | 0.865 | 0.002 | 0.151 | 0.003 |

[a]1000 replications. True values of $\beta_l$ and $\beta_k$ are 0.6 and 0.4, respectively. Standard deviations reported are of parameter estimates across the 1000 replications.

### 5.3 Table II: OP under DGP1

**Read:** Table II reports OP estimates when the DGP includes firm-specific capital adjustment costs and labor timing features.

- Regardless of materials-measurement-error level, OP is badly misspecified in this environment.
- Even with zero measurement error, OP estimates are about $\hat{\beta}_l = 0.824$ and $\hat{\beta}_k = 0.188$, far from the true values $(0.6, 0.4)$.
- As measurement error rises to 0.5, the distortion worsens slightly: $\hat{\beta}_l = 0.865$ and $\hat{\beta}_k = 0.151$.

**Interpretation.**

- The problem is not mainly measurement error in investment.
- The deeper issue is that OP's scalar-unobservable assumption is violated by firm-specific capital adjustment costs.

- This is why ACF repeatedly emphasize that LP's original insight—using a non-dynamic proxy input instead of investment—is already an important improvement over OP.

### 5.4 How to read the quantitative section overall

- The Monte Carlo section is not trying to prove a single estimator is always best.
- It is instead mapping estimators to environments.
- That mapping is the paper's practical lesson: before choosing OP, LP, ACF, Wooldridge, or dynamic panels, one must ask what the likely input-timing structure and omitted shocks look like in the industry being studied.

## 6 Short summary

- OP and LP use proxy variables to control for unobserved productivity, but both typically identify labor in a first stage that may suffer from functional dependence.
- The core problem is that labor is often a deterministic function of the same variables entering the nonparametric control function, so the first stage has no independent variation left to identify $\beta_l$.
- ACF propose inverting *conditional* rather than unconditional input demand functions.
- Their first stage only purges $\varepsilon_{it}$; all production coefficients are identified in the second stage.
- This modification allows more realistic DGPs, including dynamic labor and serially correlated unobserved wage shocks.
- In Monte Carlo experiments, LP is more efficient when its own assumptions are correct, but ACF is more robust when they are not and is generally less sensitive to measurement error in the proxy variable.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

- The paper establishes that the standard first-stage interpretation of OP and LP is too strong.
- The labor coefficient is generally *not* identified in that first stage unless one imposes special timing or noise structures.
- A closely related conditional-inversion estimator avoids this problem and extends the class of environments in which proxy-variable estimation is consistent.

### 7.2 Economic intuition

- A proxy variable helps only if it reveals productivity *without* swallowing the variation needed to identify other coefficients.

- In OP/LP, the proxy inversion often explains labor too well, leaving nothing left for labor to identify.
- ACF's solution is to stop asking the first stage to do too much. Let it recover only the control function; then let lagged instruments and productivity dynamics identify the elasticities in the second stage.

## 7.3 Takeaway

- The main lesson for applied work is methodological humility: choosing a production-function estimator is fundamentally about choosing identifying assumptions.
- ACF show that proxy-variable estimation remains useful, but only after one rethinks what the proxy inversion can actually identify.
- The paper is therefore best read as a careful identification repair of OP/LP, not as a rejection of the proxy-variable program itself.